08. Letter Sequences & Alphabet Code

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Sequence moving backwards:

What comes next in the sequence: R, Q, P, O, ?

Answer: Each letter moves back 1 place in the alphabet (R, Q, P, O = −1 each time). O − 1 = N. Answer: **N**.

Example 2 — Sequence that wraps around the alphabet:

What comes next in the sequence: U, W, Y, A, ?

Answer: Each letter moves forward 2 places, wrapping around after Z (U, W, Y, A = +2 each time, with Y+2 wrapping to A). A + 2 = C. Answer: **C**.

Example 3 — Alphabet code:

If the code for FOX is HQZ (each letter shifts forward by 2 in the alphabet), what is the code for PIG?

Answer: Shift every letter of PIG forward by 2: P→R, I→K, G→I. Answer: **RKI**.

Example 4 — Multiple-choice:

What comes next in the sequence: H, K, N, Q, ? A) R B) S C) T D) U

Answer: The gap is +3 throughout: H→K→N→Q, so Q+3 = T. Answer: **C) T**.

Example 5 — Select-all:

For the sequence H, K, N, Q, which of these are true? A) The gap between each letter is 3. B) The next letter would be T. C) The sequence goes backwards through the alphabet. D) The gap between K and N is 3. E) The first letter is H.

Answer: The gap is +3 the whole way through (A), so the next letter is Q+3=T (B); the gap between K and N is also 3 (D); H is the first letter (E). The sequence moves forward, not backward, so C is false. Answer: **A, B, D and E**.

Foundation

15 questions

1. What comes next in the sequence: T, S, R, Q, ?

2. What comes next in the sequence: V, U, T, S, ?

3. What comes next in the sequence: N, M, L, K, ?

4. What comes next in the sequence: Z, Y, X, W, ?

5. What comes next in the sequence: J, I, H, G, ?

6. If the code for BIG is CJH (each letter shifts forward by 1 in the alphabet), what is the code for RUN?

7. If the code for SUN is TVO (each letter shifts forward by 1 in the alphabet), what is the code for PEN?

8. If the code for HAT is IBU (each letter shifts forward by 1 in the alphabet), what is the code for CAP? A) DBQ B) DBR C) CBQ D) EBQ

9. What comes next in the sequence: Q, R, S, T, U, ?

10. What comes next in the sequence: F, E, D, C, ?

11. If the code for TOP is UPQ (each letter shifts forward by 1 in the alphabet), what is the code for WIN?

12. What comes next in the sequence: W, X, Y, Z, ?

13. What comes next in the sequence: M, L, K, J, ? A) G B) H C) I D) K

14. If the code for FUN is GVO (each letter shifts forward by 1 in the alphabet), what is the code for ZOO?

15. For the sequence P, R, T, V, which of these are true? A) The gap between each letter is 2. B) The next letter would be X. C) The sequence goes backwards through the alphabet. D) The gap between R and T is 2. E) The first letter is P.

Standard

20 questions

1. What comes next in the sequence: Q, S, U, W, ?

2. What comes next in the sequence: R, T, V, X, ?

3. What comes next in the sequence: Z, X, V, T, ?

4. What comes next in the sequence: Y, W, U, S, ?

5. What comes next in the sequence: S, U, W, Y, ? A) A B) Z C) B D) C

6. If the code for CAT is ECV (each letter shifts forward by 2 in the alphabet), what is the code for DOG?

7. If the code for BIRD is DKTF (each letter shifts forward by 2 in the alphabet), what is the code for FISH? A) HKUJ B) HKUI C) GKUJ D) HLUJ

8. What comes next in the sequence: G, J, M, P, ?

9. What comes next in the sequence: R, O, L, I, ?

10. For the sequence G, J, M, P, which of these are true? A) The gap between each letter is 3. B) The next letter would be S. C) The sequence goes backwards through the alphabet. D) The gap between J and M is 3. E) The first letter is G.

11. What comes next in the sequence: A, E, I, M, ?

12. What comes next in the sequence: Y, U, Q, M, ?

13. What comes next in the sequence: E, I, M, Q, ? A) S B) T C) U D) V

14. If the code for GOLD is JROG (each letter shifts forward by 3 in the alphabet), what is the code for FISH?

15. What comes next in the sequence: W, T, Q, N, K, ?

16. For the sequence G, K, O, S, which of these are true? A) The gap between each letter is 4. B) The next letter would be W. C) The sequence goes backwards through the alphabet. D) The gap between K and O is 4. E) The first letter is G.

17. What comes next in the sequence: V, T, R, P, ?

18. If the code for STAR is UVCT (each letter shifts forward by 2 in the alphabet), what is the code for MOON?

19. What comes next in the sequence: U, R, O, L, ? A) G B) H C) I D) J

20. What comes next in the sequence: T, V, X, Z, ?

Hard

5 questions

1. What comes next in the sequence: B, G, L, Q, ?

2. What comes next in the sequence: Z, U, P, K, ? A) E B) F C) G D) H

3. For the sequence B, G, L, Q, which of these are true? A) The gap between each letter is 5. B) The next letter would be V. C) The sequence goes backwards through the alphabet. D) The gap between G and L is 5. E) The first letter is B.

4. If the code for WAVE is AEZI (each letter shifts forward by 4 in the alphabet, wrapping round where needed), what is the code for PLUM?

5. What comes next in the sequence: S, V, Y, B, ?

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